In [5]:

```
import numpy as np

P = np.array([
    [0.00, 0.50, 0.40, 0.10],
    [0.40, 0.00, 0.40, 0.20],
    [0.35, 0.50, 0.00, 0.15],
    [0.00, 0.00, 0.00, 1.00],
])

def estimate_exit_probability(N, n_sims=100_000, seed=None):

    if seed is not None:
        np.random.seed(seed)

    states = np.zeros(n_sims, dtype=np.int8)

    for _ in range(N):
        for s in (0, 1, 2):
            mask = (states == s)
            count = mask.sum()
            if count > 0:
                states[mask] = np.random.choice(4, size=count, p=P[s])

    return (states == 3).mean()

if __name__ == "__main__":
    Ns = [5, 10, 20, 30]
    n_sims = 200_000
    seed = 50

    print(f"Simulations per N = {n_sims}, seed = {seed}\n")
    for N in Ns:
        prob = estimate_exit_probability(N, n_sims=n_sims, seed=seed)
        print(f"After N = {N:2d} steps: estimated P(exit) = {prob:.5f}")
```

```
Simulations per N = 200000, seed = 50

After N =  5 steps: estimated P(exit) = 0.80729
After N = 10 steps: estimated P(exit) = 0.96267
After N = 20 steps: estimated P(exit) = 0.99869
After N = 30 steps: estimated P(exit) = 0.99992
```

In [ ]: